Breast Cancer Detection

Group Members

Name	Roll Number	Role
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What We Are Trying to Do, and Why

We classify mammogram (breast X-ray) images as benign or malignant using a transfer-learning convolutional neural network alongside classical classifiers trained on handcrafted image features (texture and statistical descriptors), and compare their performance. The model informs a computer-aided detection (CAD) style second-opinion tool for radiologists – flagging mammograms likely to be malignant for priority review, not replacing diagnosis. This matters because manual mammogram screening is time-consuming and subject to human variability; an assistive triage signal can help prioritise cases in high-volume or resource-limited screening settings, supporting faster, more consistent review.

Dataset / Problem Source

Dataset: The Complete Mini-DDSM (JPEG-8 subset) – Kaggle: kaggle.com/datasets/cheddad/miniddsm2, derived from the original DDSM mammography collection.
Licence: CC Attribution-NoDerivatives 4.0 International. Cite as: C.D. Lekamlage, F. Afzal, E. Westerberg and A. Cheddad, "Mini-DDSM: Mammography-based Automatic Age Estimation," DMIP 2020, ACM, pp. 1–6, doi:10.1145/3441369.3441370; original DDSM: Heath et al., IWDM 2001.
Size: 2,684 benign and 2,716 malignant usable mammogram images (5,400 total; Normal class excluded, not used for this binary classification task), confirmed by direct file count.

Tools & Technologies

Language/Libraries: Python; TensorFlow/Keras (or PyTorch), OpenCV, scikit-learn, Pandas, Matplotlib (run on Google Colab, free-tier GPU/CPU).
Planned Models: Transfer-learning CNN (pretrained ResNet/EfficientNet, fine-tuned classification head) as the deep-learning model; SVM and Random Forest trained on handcrafted texture/statistical image features as classical models – at least two model families, deep learning used alongside classical models, not alone. Majority-class baseline for comparison. Train/test split performed at patient level to prevent leakage across multiple views of the same case.
GUI/Deployment: Flask web interface accepting an uploaded mammogram image and returning a live benign/malignant prediction.